
Heaven's Light Is Our Guide



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

COURSE TITLE

Digital Techniques Sessional

Course No. ECE 2112

Lab No. 03

REPORT TITLE

**Simplification and Simulation of
Boolean Expressions Using Logic Gates**

Date of Submission: 23 August, 2026

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Problem: 01

Problem Definition:

Simplification and implementation of the Boolean expression

$$F = AB + \overline{B}C + AC + \overline{A}B\overline{D}$$

using Boolean algebra and verification of the resulting logic circuit through simulation.

Objective:

To simplify the given four-variable Boolean expression using Boolean algebra, obtain its minimized form, and verify the original and simplified circuits using logic simulation.

Algebraic Simplification:

$$\begin{aligned} F &= AB + \overline{B}C + AC + \overline{A}B\overline{D} \\ &= AB + \overline{B}C + AC + \overline{A}B\overline{D} \\ &= AB + \overline{B}C + AC(B + \overline{B}) + \overline{A}B\overline{D} \\ &= AB + \overline{B}C + ABC + \overline{B}AC + \overline{A}B\overline{D} \\ &= AB + \overline{B}C + \overline{A}B\overline{D} \end{aligned}$$

Therefore,

$$F = AB + \overline{B}C + B\overline{D}$$

Truth Table:

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Table 1: Truth table for $F = AB + \overline{B}C + B\overline{D}$

Karnaugh Map:

$AB \backslash CD$	00	01	11	10
00	0	0	1	1
01	1	0	0	1
11	1	1	1	1
10	0	0	1	1

Table 2: K-map for $F = AB + \overline{B}C + AC + \overline{A}B\overline{D}$

The K-map groups can be selected to obtain the minimized expression $AB + \overline{B}C + B\overline{D}$. The term AC is redundant because its 1-cells are already covered by the other groups.

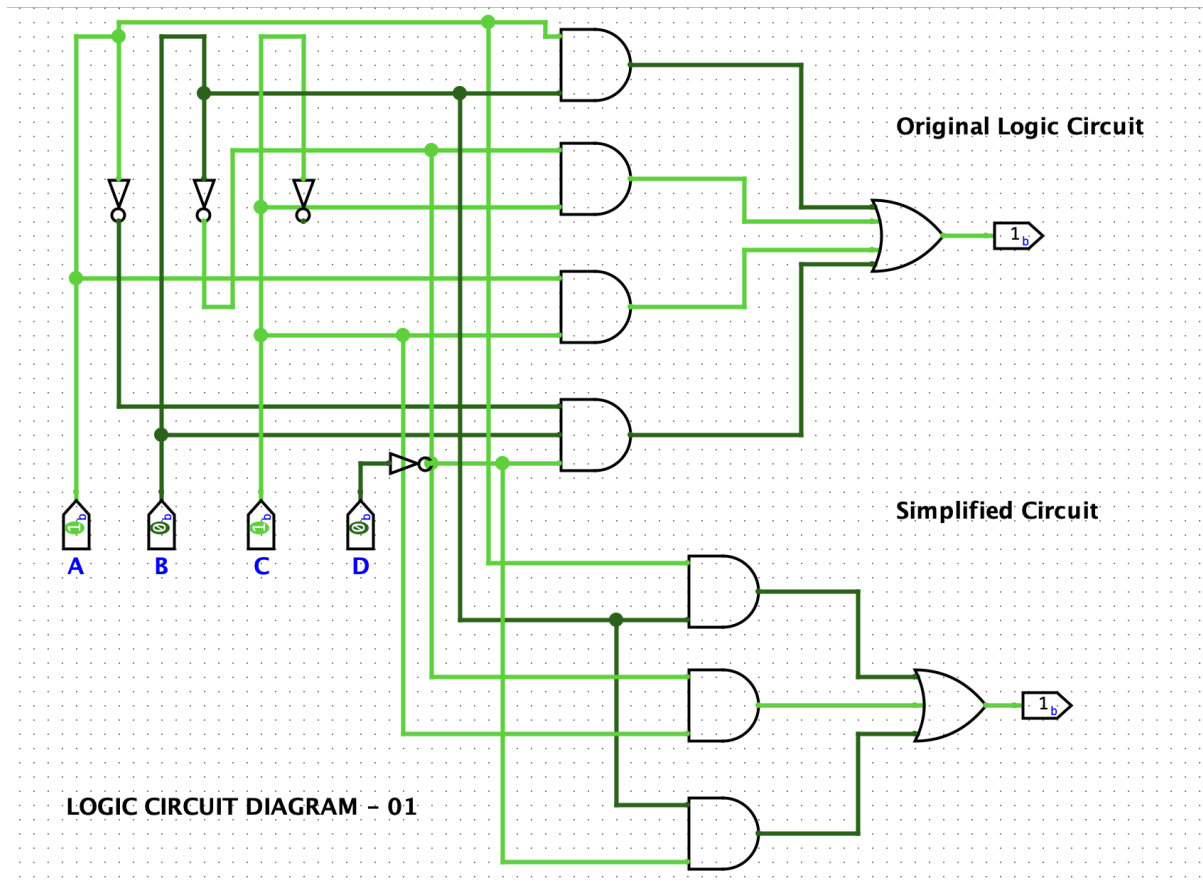
Circuit Diagram:

Figure 1: Original and simplified logic circuits for Problem 1, simulated in Logisim.

Discussion:

The original circuit directly implements all four product terms of the given expression. After simplification, the term AC becomes redundant, reducing the number of required logic gates. The simplified circuit therefore provides the same output with a more efficient

arrangement of logic gates. The simulation verifies that both circuits produce identical outputs for all possible input combinations.

Conclusion:

The given Boolean expression was successfully simplified to $F = AB + \overline{B}C + B\overline{D}$. The result was verified using the Karnaugh map and logic circuit simulation.

Problem: 02

Problem Definition:

Simplification of the Boolean expression

$$F = \overline{(\overline{A} + B)(C + \overline{D})} [A\overline{B} + \overline{(\overline{C} \overline{D})}]$$

using Boolean algebra and verification through logic simulation.

Objective:

To apply De Morgan's theorem and Boolean algebra to simplify the given expression and verify the simplified result using a logic circuit simulation.

Algebraic Simplification:

$$\begin{aligned} F &= \overline{(\overline{A} + B)(C + \overline{D})} [A\overline{B} + \overline{(\overline{C} \overline{D})}] \\ &= \left[\overline{(\overline{A} + B)} + \overline{(C + \overline{D})} \right] [A\overline{B} + CD] \\ &= (A\overline{B} + \overline{C}D)(A\overline{B} + CD) \\ &= A\overline{B} + (\overline{C}D)(CD) \\ &= A\overline{B} \end{aligned}$$

Therefore,

$$\boxed{F = A\overline{B}}$$

Truth Table:

Karnaugh Map:

The four 1-cells corresponding to $A = 1$ and $B = 0$ form a group of four. The variables C and D are eliminated, resulting in $F = A\overline{B}$.

Circuit Diagram:

Discussion:

A	B	C	D	$F = A\overline{B}$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Table 3: Truth table for $F = A\overline{B}$

$AB \backslash CD$	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	0
10	1	1	1	1

Table 4: K-map for the simplified expression $F = A\overline{B}$

De Morgan's theorem converts the complemented product into a sum of complemented terms. The resulting expression contains two factors with a common term $A\overline{B}$. Applying the distributive identity $(X + Y)(X + Z) = X + YZ$ reduces the expression to $A\overline{B}$. The simulation confirms that the simplified circuit produces the same output as the original circuit.

Conclusion:

The given expression was successfully reduced to $F = A\overline{B}$ using De Morgan's theorem and Boolean algebra. The Karnaugh map and logic simulation verified the correctness of the simplification.

Problem: 03

Problem Definition:

Simplification of the four-variable Boolean expression

$$F = ABCD + A\overline{B}CD + \overline{A}BCD + AB\overline{C}D + ABC\overline{D}$$

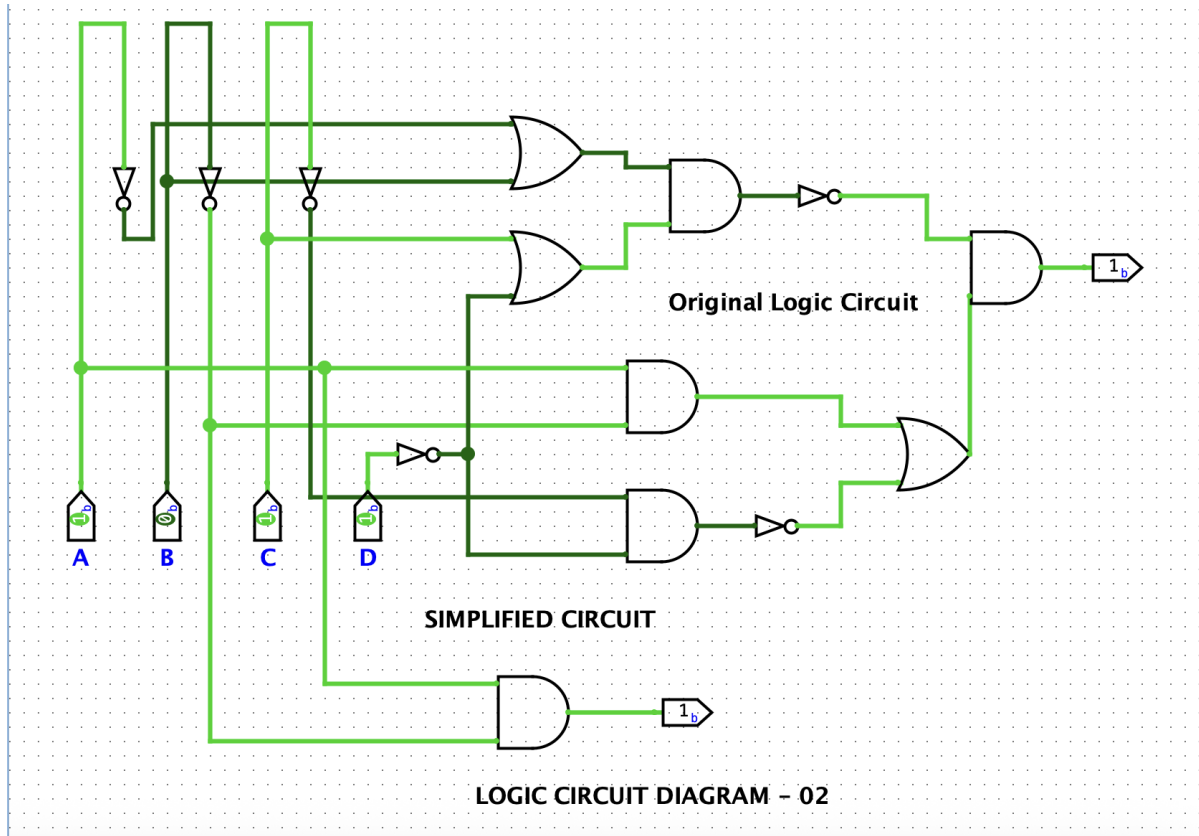


Figure 2: Original and simplified logic circuits for Problem 2, simulated in Logisim.

using Boolean algebra and Karnaugh map.

Objective:

To minimize the given sum-of-products Boolean expression and verify the result using a Karnaugh map and logic circuit simulation.

Algebraic Simplification:

$$\begin{aligned}
 F &= ABCD + A\bar{B}CD + \bar{A}BCD + AB\bar{C}D + ABC\bar{D} \\
 &= ABC(D + \bar{D}) + ABD(C + \bar{C}) + ACD(B + \bar{B}) + BCD(A + \bar{A}) \\
 &= ABC + ABD + ACD + BCD
 \end{aligned}$$

Therefore,

$$F = ABC + ABD + ACD + BCD$$

Truth Table:

Karnaugh Map:

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Table 5: Truth table for $F = ABC + ABD + ACD + BCD$

$AB \backslash CD$	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	1	1	1
10	0	0	1	0

Table 6: K-map for $F = ABCD + \overline{A}BCD + \overline{A}BC\overline{D} + \overline{A}B\overline{C}D + \overline{A}B\overline{C}\overline{D}$

The K-map allows four groups of two adjacent cells. These groups produce the terms ABC , ABD , ACD , and BCD . Therefore, the minimized expression is

$$F = ABC + ABD + ACD + BCD.$$

Simplified Expression from Logisim:

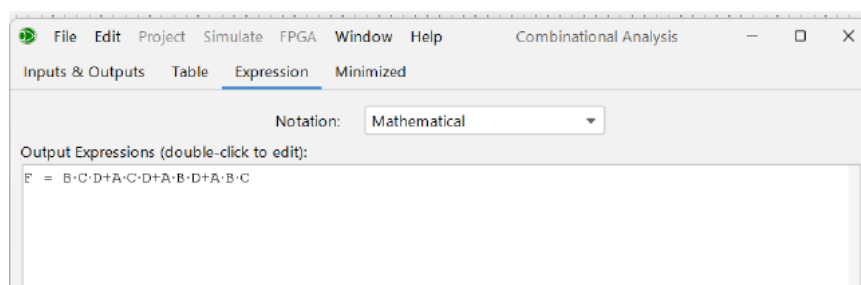


Figure 3: Simplified Boolean expression obtained from Logisim for Problem 3.

Simplified Circuit from Logisim:

Discussion:

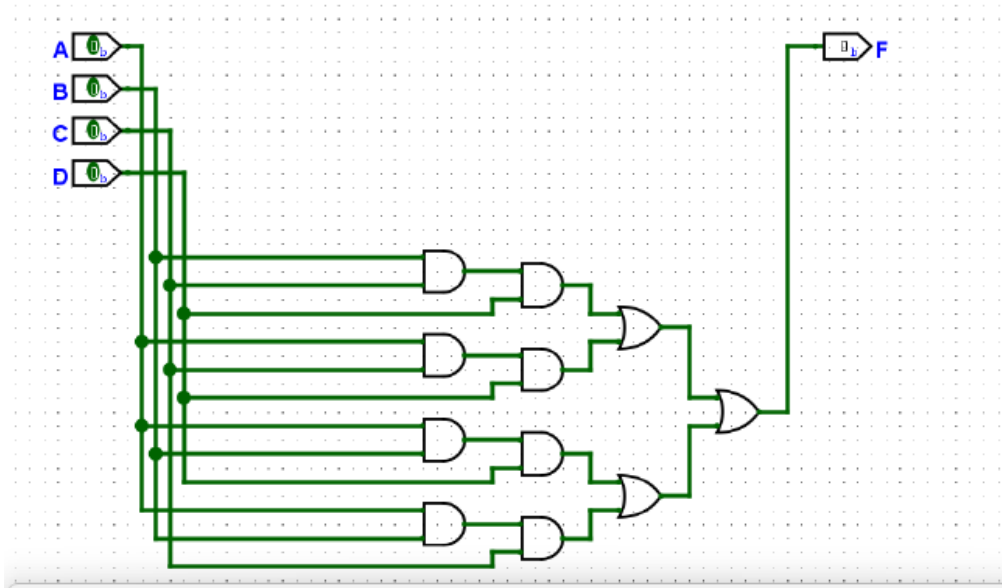


Figure 4: Simplified logic circuit for Problem 3 implemented in Logisim.

The Karnaugh map was used to minimize the given Boolean expression. The simplified expression obtained was $F = ABC + ABD + ACD + BCD$. The simplified expression was then implemented in Logisim using the required logic gates. The Logisim simulation confirms the correctness of the simplified circuit.

Conclusion:

The given four-variable Boolean expression was successfully minimized to $F = ABC + ABD + ACD + BCD$. The simplification was verified using the Karnaugh map and the corresponding simplified logic circuit was successfully implemented in Logisim.

Problem: 04

Problem Definition:

Simplification of the product-of-sums Boolean expression

$$F = (A + B + C + D)(A + B + \overline{C})(A + \overline{B} + C)(\overline{A} + B + C)$$

using Boolean algebra and Karnaugh map.

Objective:

To simplify the given product-of-sums expression, determine its minimal sum-of-products form, and verify the result using a Karnaugh map and logic circuit simulation.

Algebraic Simplification:

The four maxterms correspond to the zero locations

$$F = \Pi M(1, 2, 4, 7).$$

Therefore, the remaining minterms are

$$F = \Sigma m(0, 3, 5, 6, 8, 9, 10, 11, 12, 13, 14, 15).$$

Using Karnaugh-map grouping, the simplified expression is

$$F = AB + AC + BC + \overline{A}\overline{B}\overline{C}$$

Truth Table:

A	B	C	D	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Table 7: Truth table for Problem 4

Karnaugh Map:

$AB \backslash CD$	00	01	11	10
00	1	0	1	0
01	0	1	0	1
11	1	1	1	1
10	0	1	1	1

Table 8: K-map for Problem 4

The groups corresponding to AB , AC , and BC cover the majority function region. The isolated group corresponding to $A = 0$, $B = 0$, $C = 0$ gives the term $\overline{A}\overline{B}\overline{C}$. Hence,

$$F = AB + AC + BC + \overline{A}\overline{B}\overline{C}.$$

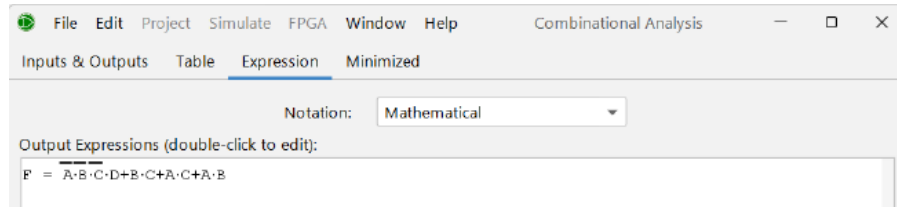


Figure 5: Simplified Boolean expression obtained from Logisim for Problem 4.

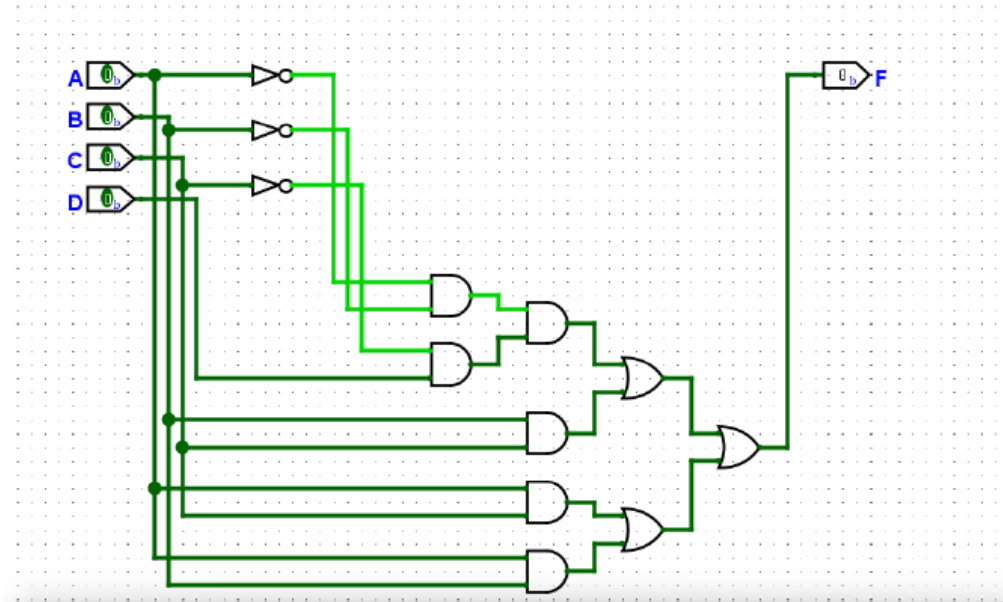


Figure 6: Simplified logic circuit for Problem 4 implemented in Logisim.

Simplified Expression from Logisim:

Simplified Circuit from Logisim:

Discussion:

The Karnaugh map was used to obtain the simplified sum-of-products expression $F = AB + AC + BC + \overline{ABC}$. The simplified expression was then implemented in Logisim using the required logic gates. The simulation confirms the correctness of the simplified circuit.

Conclusion:

The given product-of-sums Boolean expression was successfully simplified to

$$F = AB + AC + BC + \overline{ABC}.$$

The result was verified using the Karnaugh map and the corresponding simplified logic circuit was successfully implemented in Logisim.

Problem: 05

Problem Definition:

Simplification of the Boolean expression

$$F = [A + \overline{B}(C + D)][\overline{A} + B(\overline{C} + \overline{D})]$$

using Boolean algebra and Karnaugh map.

Objective:

To simplify the given Boolean expression containing nested AND and OR operations, obtain its minimal form, and verify the result using a Karnaugh map and logic circuit simulation.

Algebraic Simplification:

Let

$$X = A + \overline{B}(C + D)$$

and

$$Y = \overline{A} + B(\overline{C} + \overline{D}).$$

Then,

$$\begin{aligned} F &= [A + \overline{B}(C + D)][\overline{A} + B(\overline{C} + \overline{D})] \\ &= AB(\overline{C} + \overline{D}) + \overline{A}\overline{B}(C + D) \\ &\quad + \overline{B}(C + D)B(\overline{C} + \overline{D}) \\ &\quad + A\overline{A} \\ &= AB\overline{C} + AB\overline{D} + \overline{A}\overline{B}C + \overline{A}\overline{B}D \end{aligned}$$

since

$$A\overline{A} = 0$$

and

$$\overline{B}B = 0.$$

Therefore,

$$\boxed{F = AB\overline{C} + AB\overline{D} + \overline{A}\overline{B}C + \overline{A}\overline{B}D}$$

Truth Table:

Karnaugh Map:

A	B	C	D	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Table 9: Truth table for Problem 5

$AB \backslash CD$	00	01	11	10
00	0	1	1	1
01	0	0	0	0
11	1	1	0	1
10	0	0	0	0

Table 10: K-map for Problem 5

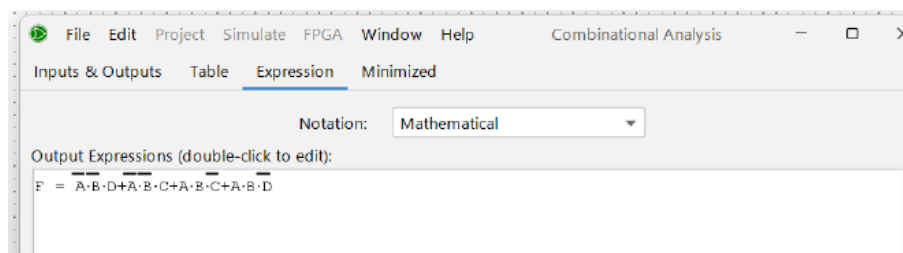


Figure 7: Simplified Boolean expression obtained from Logisim for Problem 5.

The 1-cells are grouped according to the terms $ABC\bar{D}$, $AB\bar{D}$, $\bar{A}BC$, and $\bar{A}BD$. Therefore, the minimized expression is

$$F = ABC\bar{D} + AB\bar{D} + \bar{A}BC + \bar{A}BD.$$

Simplified Expression from Logisim:

Simplified Circuit from Logisim:

Discussion:

The given Boolean expression was simplified using Boolean algebra and the Karnaugh

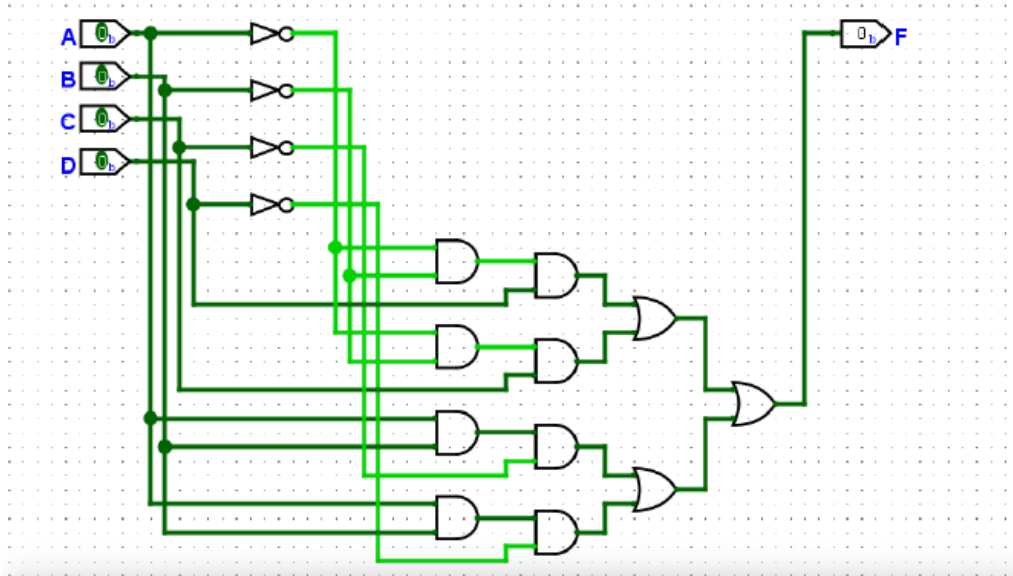


Figure 8: Simplified logic circuit for Problem 5 implemented in Logisim.

map. The resulting expression contains four product terms. The simplified expression was implemented in Logisim using the required logic gates. The simulation confirms the correctness of the simplified circuit.

Conclusion:

The given Boolean expression was successfully simplified to

$$F = ABC\bar{C} + AB\bar{D} + \bar{A}BC + \bar{A}\bar{B}D.$$

The simplification was verified using Boolean algebra, the Karnaugh map, and the corresponding simplified logic circuit implemented in Logisim.

References

1. M. M. Mano, *Digital Design*, 5th ed. Upper Saddle River, N.J.: Pearson, 2013.
2. R. J. Tocci, N. S. Widmer, and G. L. Moss, *Digital Systems: Principles and Applications*, 11th ed. Upper Saddle River, N.J.: Pearson, 2011.